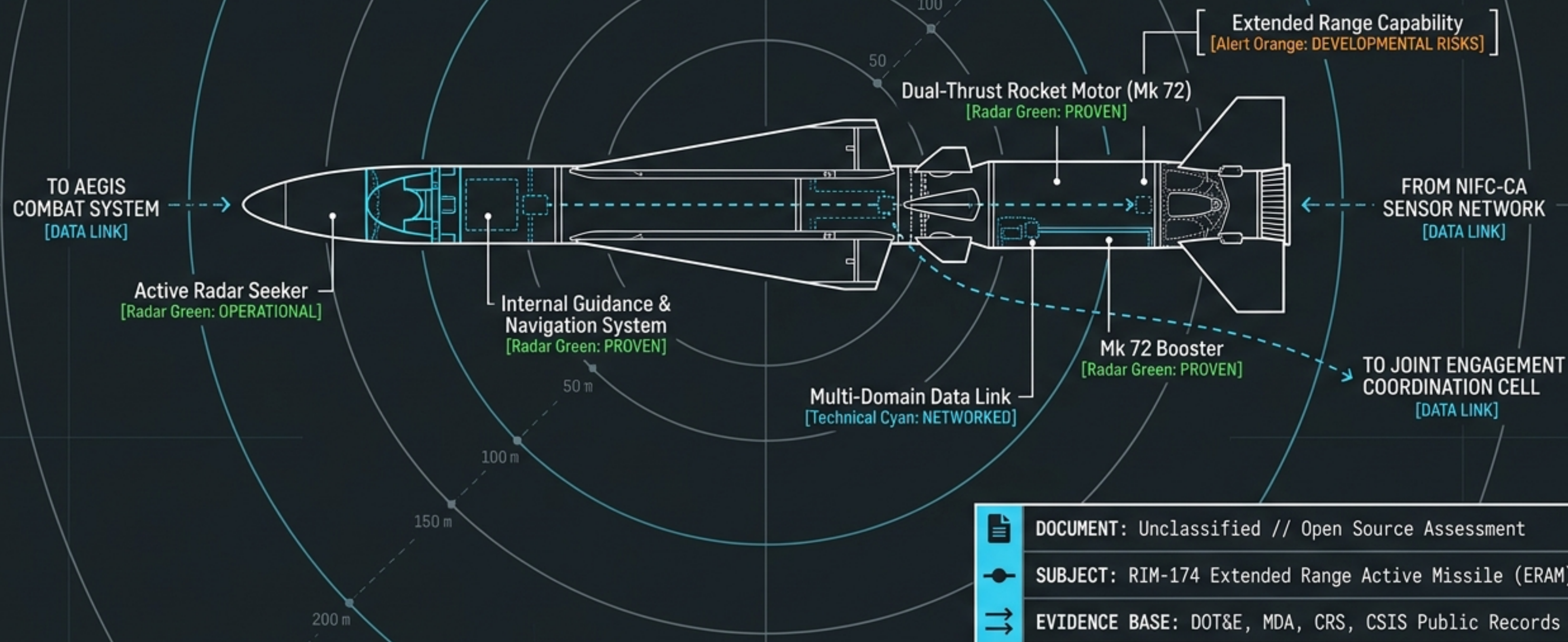


Raytheon SM-6: Capabilities, Architecture, and Reality

An evidence-graded enterprise assessment of the U.S. military's most versatile—and heavily networked—multi-domain effector.



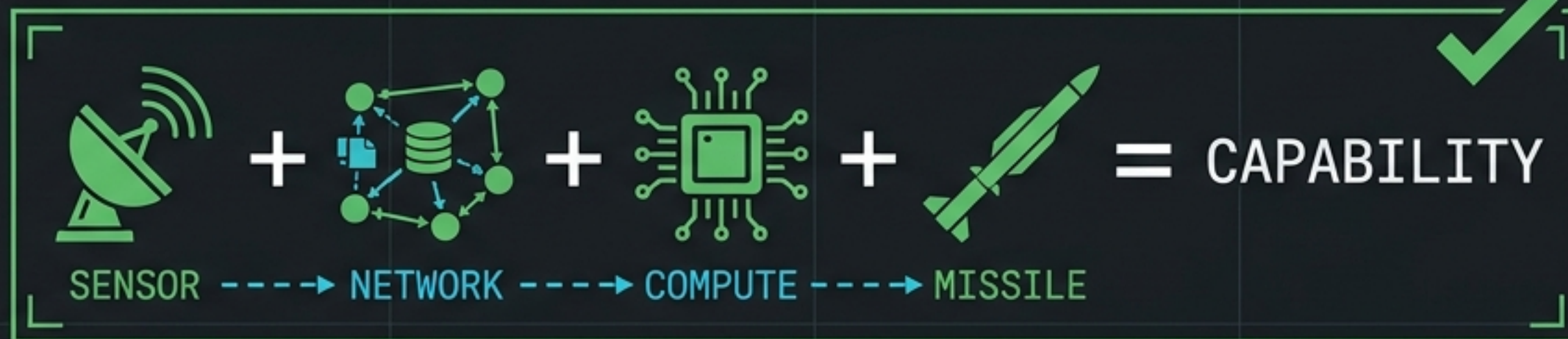
THE SM-6 IS NOT A MISSILE; IT IS AN ENDPOINT

THE SM-6 IS A HIGH-VALUE, MULTI-DOMAIN EFFECTOR WHOSE PERFORMANCE SCALES ENTIRELY WITH THE RESILIENCE AND QUALITY OF THE KILL WEB AROUND IT.

THE FLAWED MODEL



THE ARCHITECTURAL REALITY



SCALABLE ARCHITECTURE

NETWORK-DEPENDENT REACH

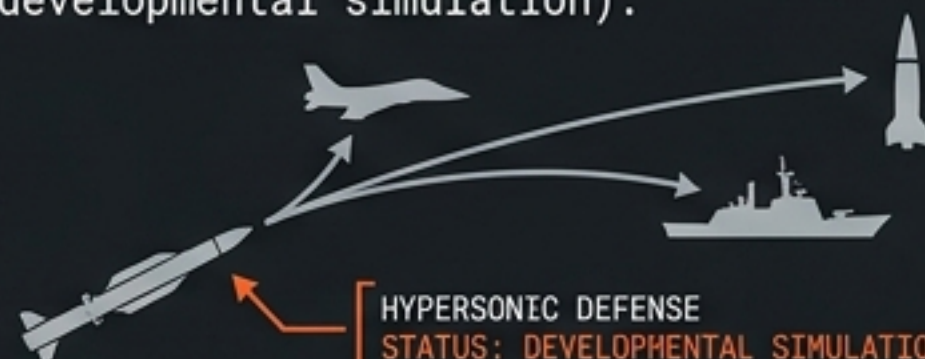
[OVER-THE-HORIZON CAPABILITY] relies on remote sensors (E-2D, Space) and networks (CEC, IBCS), not just missile kinematics.

[DATA LINK: CEC/IBCS
SENSOR INPUT: EXTERNAL (E-2D, SPACE)]



MULTI-MISSION UTILITY

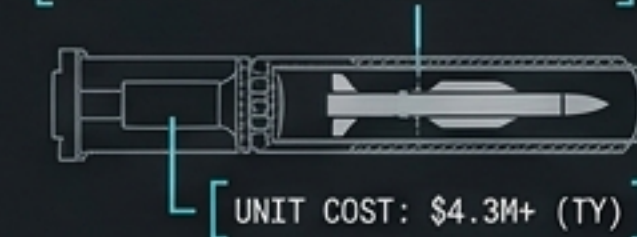
One airframe executes [EXTENDED AIR DEFENSE], [TERMINAL BMD], and [ANTI-SURFACE WARFARE]. (Hypersonic developmental simulation).



STRICT MAGAZINE ECONOMICS

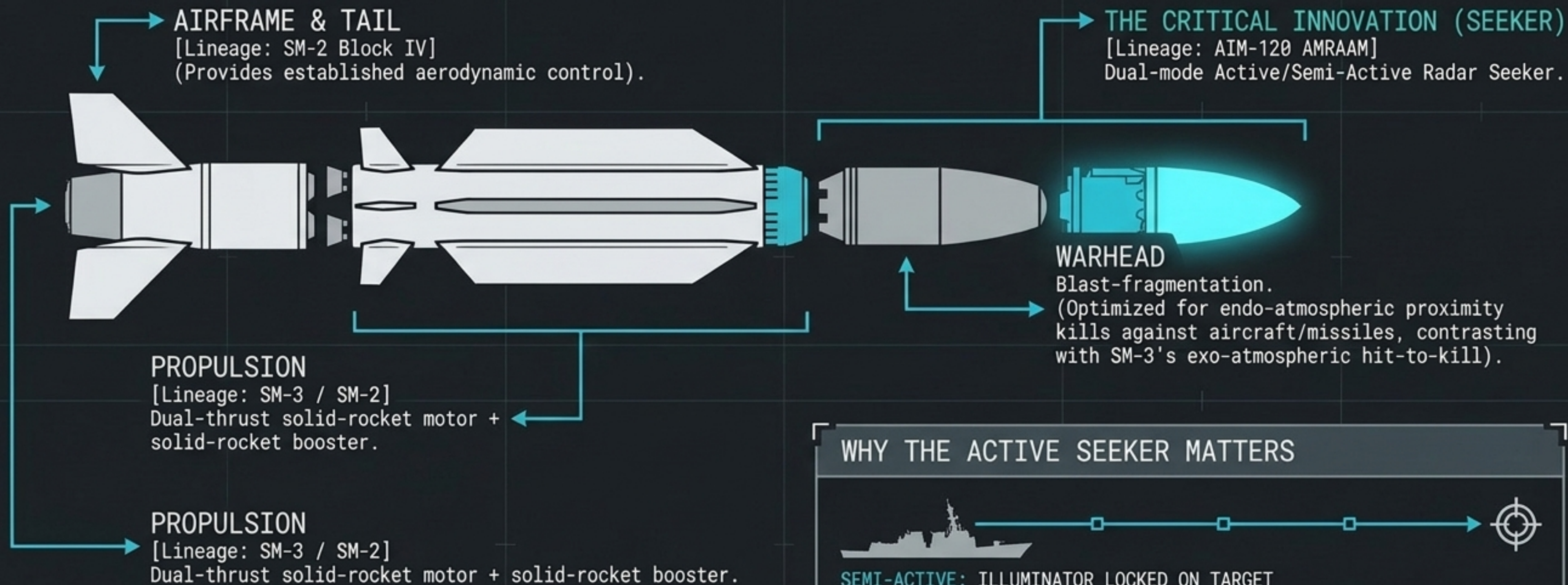
Premium interceptor (\$4.3M+ then-year cost) consuming a full [Mk 41 VLS CELL]. Firing doctrine must prioritize hard targets over cheap drones.

[Mk 41 VLS CELL CAPACITY: 1 SM-6]

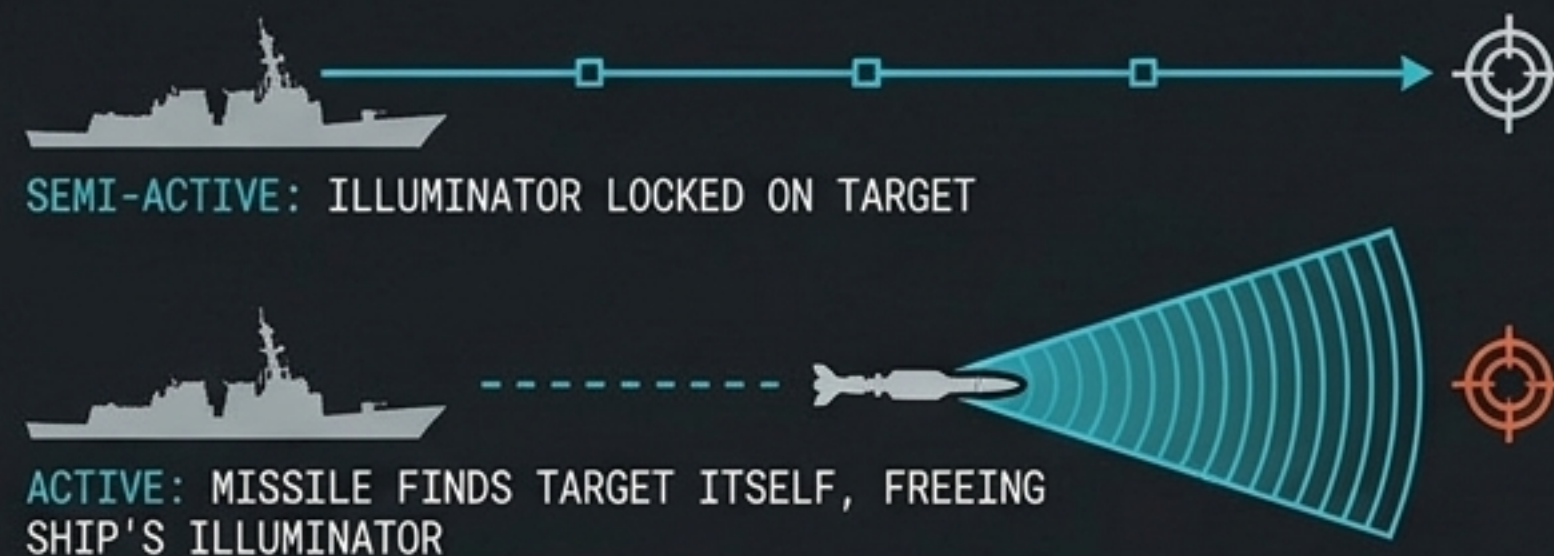


[UNIT COST: \$4.3M+ (TY)]

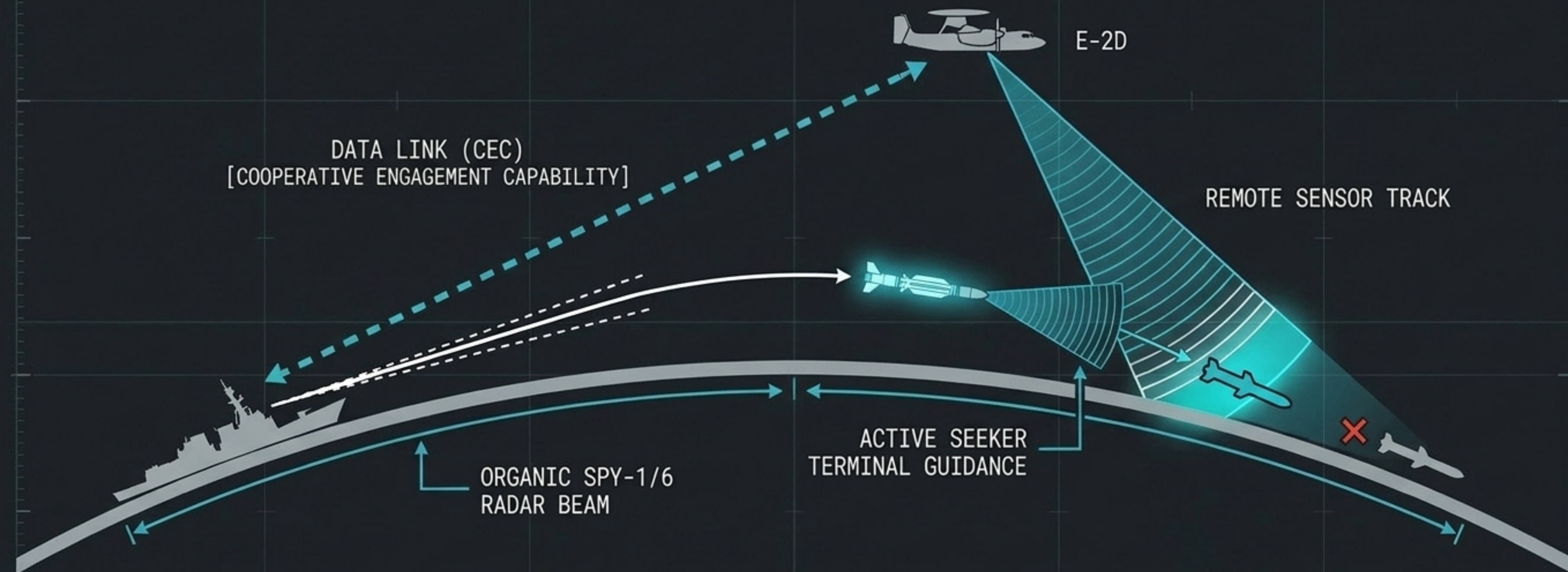
THE ANATOMY OF RIM-174: A SYNTHESIS OF LEGACY SYSTEMS



WHY THE ACTIVE SEEKER MATTERS

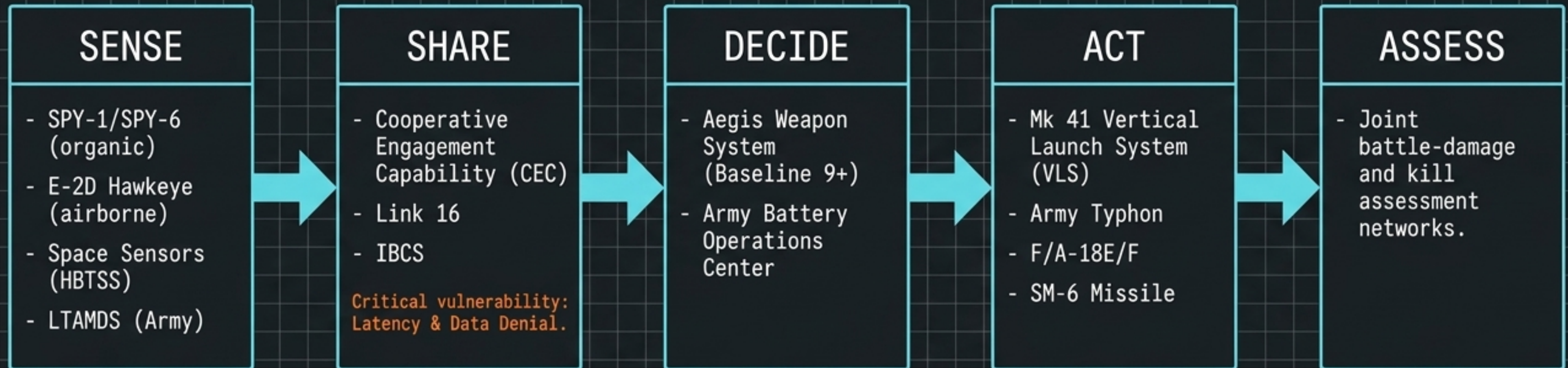


THE CORE PARADIGM SHIFT: SEPARATION OF SENSOR AND SHOOTER



ANALYTICAL TAKEAWAY: THE MISSILE'S ADVERTISED RANGE IS OPERATIONALLY IRRELEVANT IF THE FORCE CANNOT ESTABLISH AND MAINTAIN A VALID, FIRE-CONTROL-QUALITY REMOTE TRACK.

THE KILL WEB ARCHITECTURE: A SYSTEM OF SYSTEMS



The weakest link in this chain bounds the **total capability of the SM-6.**

A Multi-Mission Effector Portfolio



[Proven]

Extended-Range Air Warfare (AAW)

- Target: Aircraft, UAVs, Cruise Missiles.
- Note: Enables 'Engage-on-Remote' via NIFC-CA architecture.



[Proven]

Terminal Ballistic-Missile Defense (SBT)

- Target: Short/Medium-Range Ballistic Missiles in their final phase.
- Note: Live intercept demonstrated (FTM-32, 2024).



[Demonstrated]

Anti-Surface Warfare (ASuW)

- Target: Maritime surface combatants.
- Note: Fast kinematic strike, though blast-frag warhead produces different damage mechanisms than heavy, purpose-built anti-ship missiles.



[Developmental / Simulated]

Counter-Hypersonic

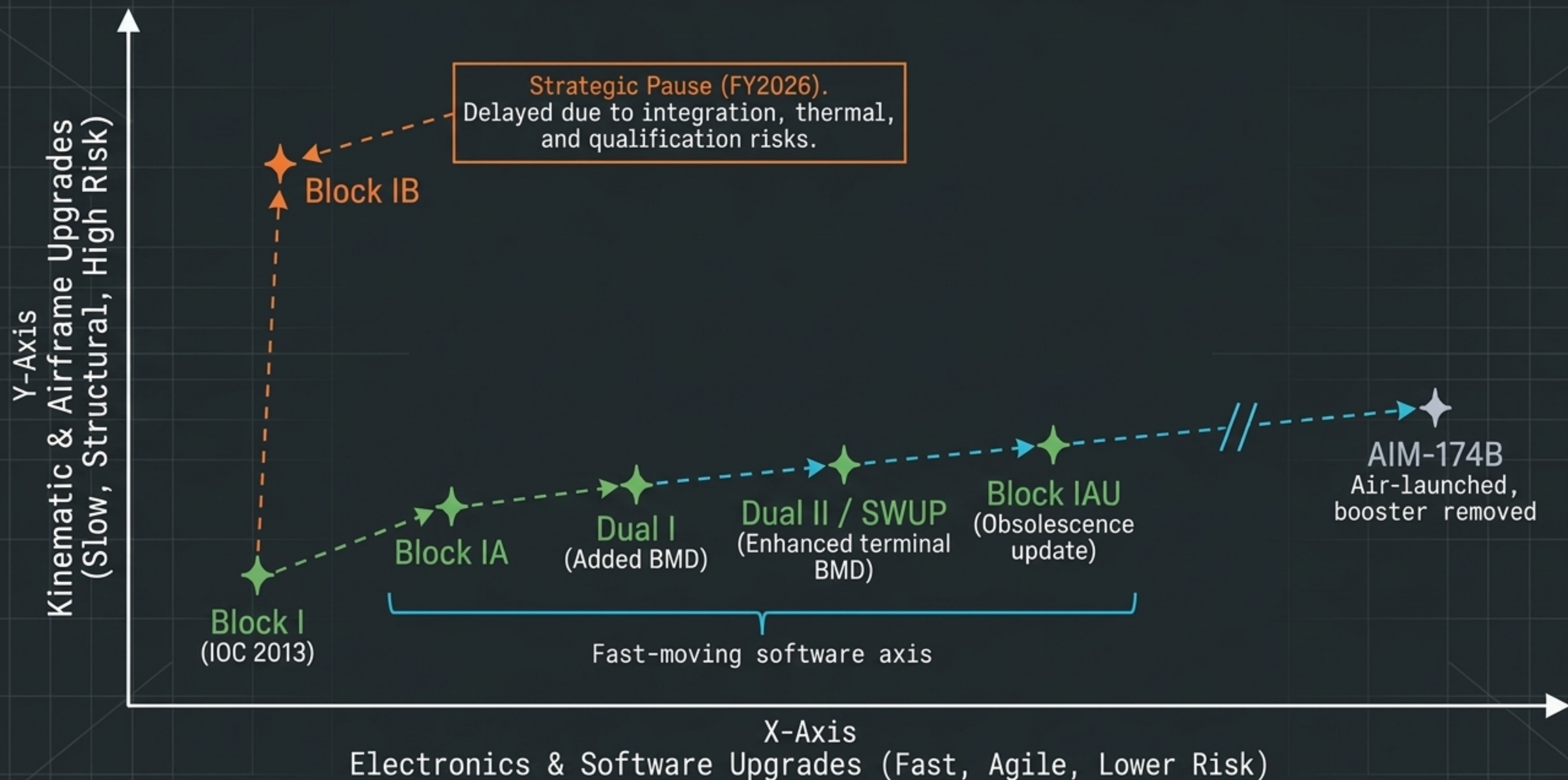
- Note: Limited to the terminal layer. 2025 FTX-40 event demonstrated tracking and simulated engagement only; live intercept remains pending.

Where SM-6 Fits: The Layered Defense Matrix

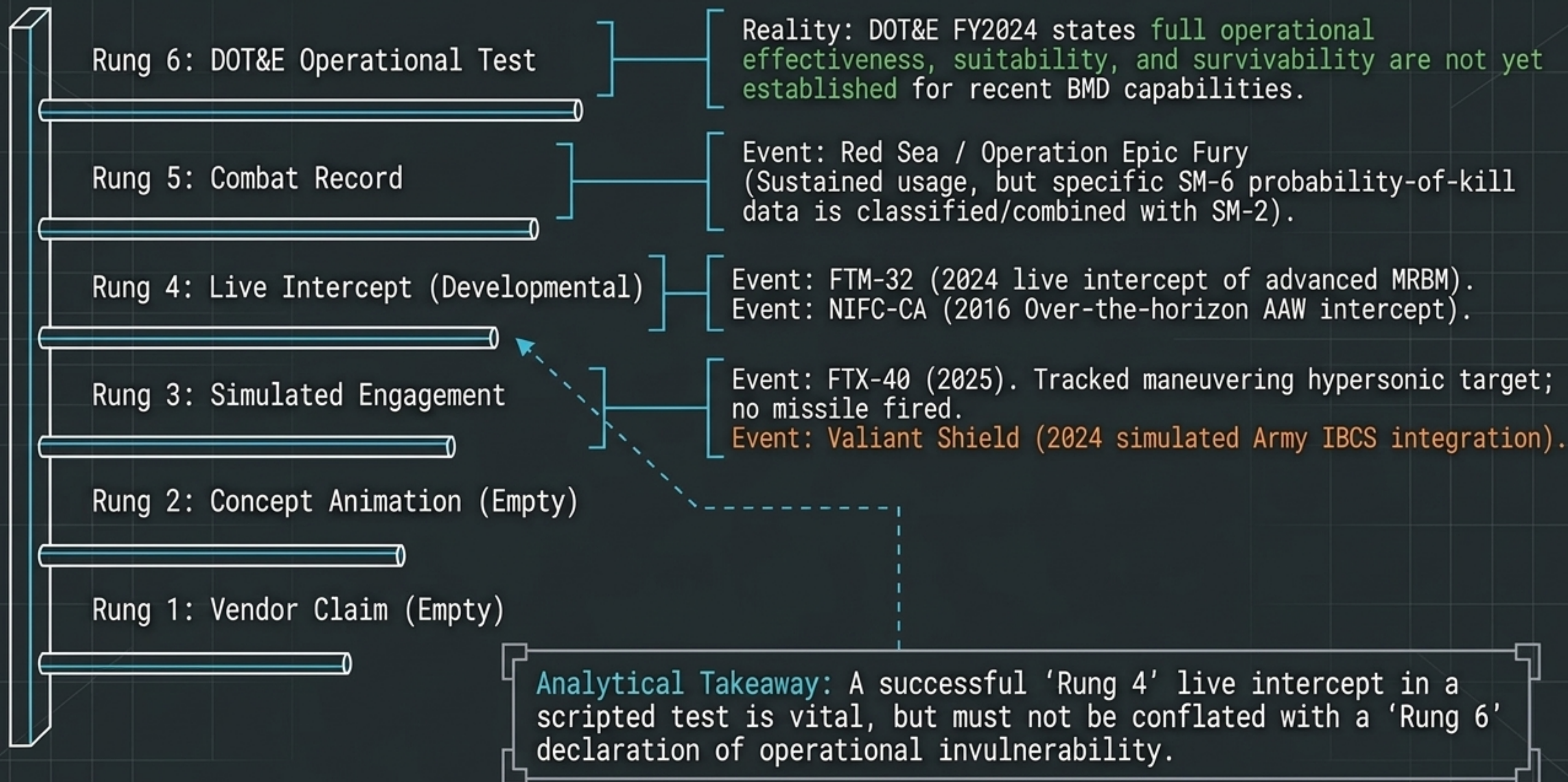


Synthesis Note: Using an SM-6 against a close-in threat wastes a premium terminal-BMD asset that could have been handled by an ESSM. SM-6 is not a substitute for SM-3's broad midcourse defense.

The Variant Roadmap: Separating Software Speed from Hardware Friction



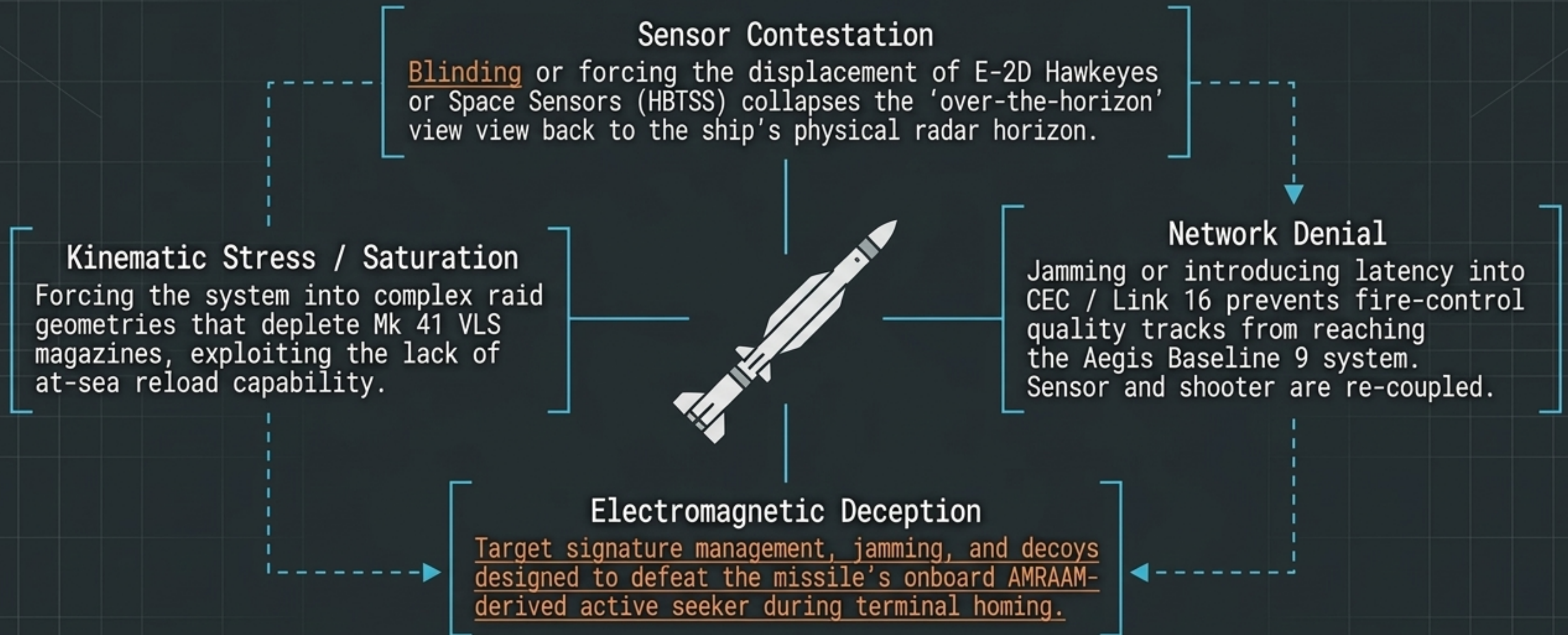
The Evidence Ladder: Grading the Public Test Record



Claim vs. Reality: Correcting the Public Narrative

Public Claim	Demonstrated Event	The DOT&E / MDA Reality
SM-6 is proven against hypersonic missiles.	FTX-40 (March 2025).	False as stated. FTX-40 was a simulated terminal engagement to reduce risk. A live hypersonic intercept (e.g., FTM-43) is not public.
SM-6 has a range of 460 km.	Various unofficial press estimates.	Official kinematic range and engagement envelopes are strictly classified.
Recent intercepts prove SM-6 is fully operationally effective.	FTM-32 (2024 MRBM Intercept).	DOT&E reported that FTM-32 was not designed to determine operational effectiveness, lethality, or survivability.

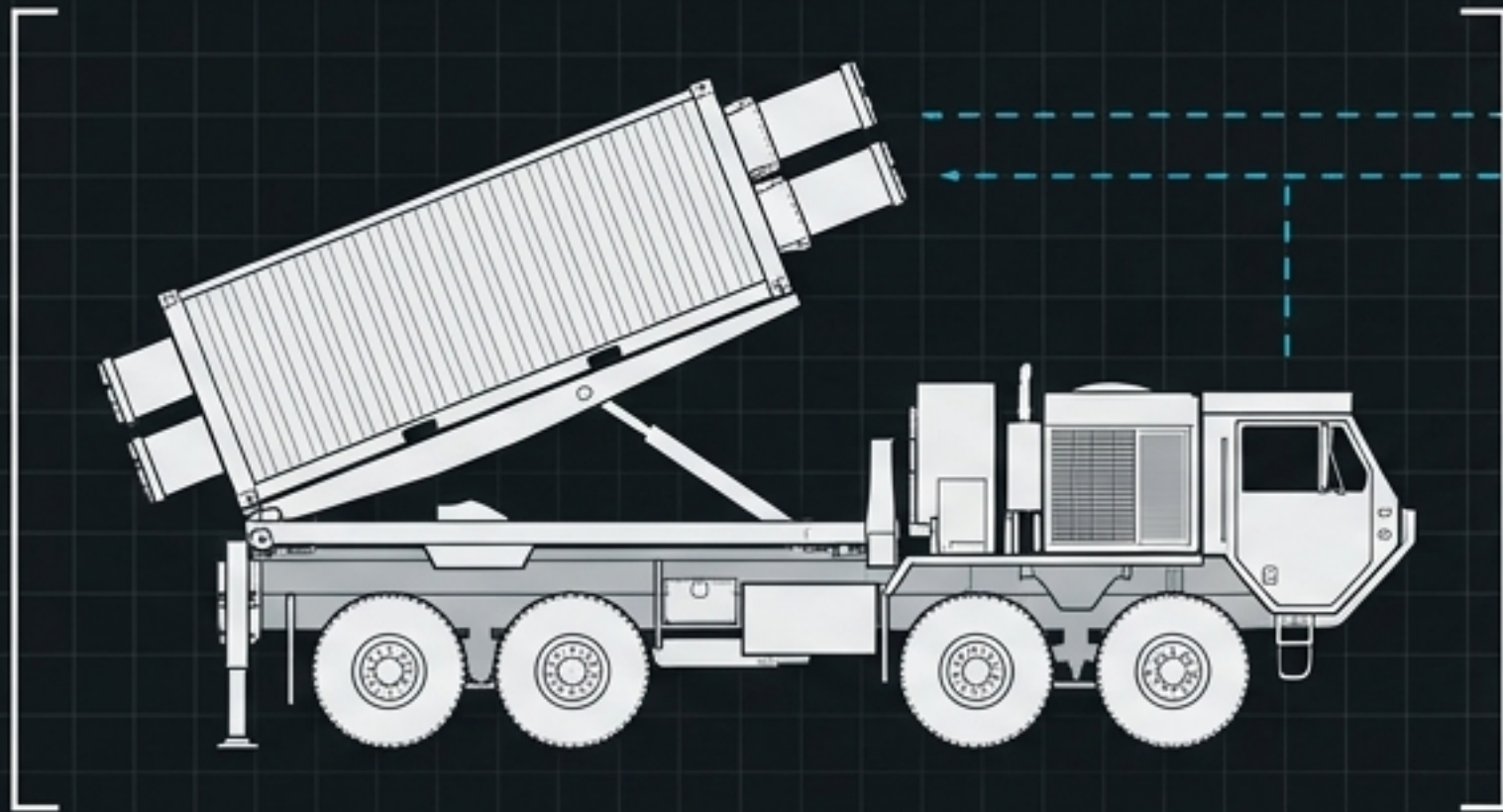
Countermeasure Vulnerabilities: Breaking the Sensor-to-Shooter Chain



Bottom Line: DOT&E notes insufficient public data to quantify SM-6 survivability in a highly contested electromagnetic/cyber environment. **Uncertainty is the correct analytical baseline.**

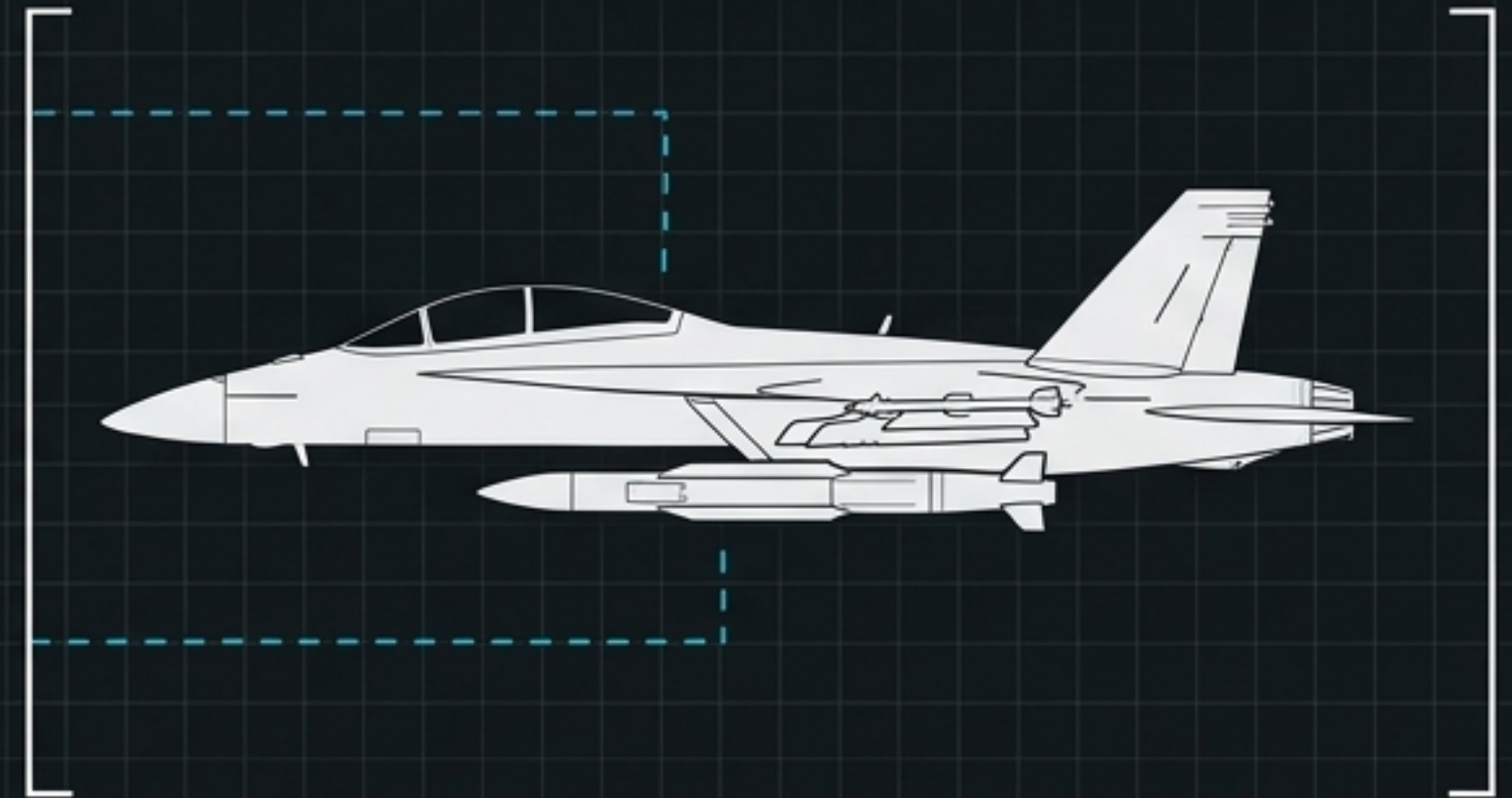
Joint Expansion: Breaking the Naval VLS Constraint

Land Domain (Army Typhon / MRC)



- Containerized, road-mobile Mk 41 derivative.
- Deployed to Philippines (April 2024).
- Conducted a maritime strike sinking exercise (Talisman Sabre 2025).
- Transforms a fleet defense round into an expeditionary sea-denial asset.

Air Domain (AIM-174B Gunslinger)



- Operationally deployed.
- Strips the heavy surface-launch booster.
- Uses aircraft kinematic energy to extend the SM-6 active-seeker technology deep into the air-to-air domain.

Magazine Economics & Strategic Signals to Watch

The VLS Constraint



[ONE CELL =
ONE ROUND]



ESSM (4-Pack)
[ONE CELL =
FOUR ROUNDS]

The Economic Reality: Cost-exchange ratios are complex (SM-6 vs a \$2K drone looks bad, but SM-6 vs a \$2B destroyer looks good). The true constraint is capacity. Every SM-6 fired at a **low-tier threat** is a **VLS cell unavailable for Terminal BMD**.

Procurement Cost: ~\$2.97M (constant 2004 base) vs. ~\$4.3M (then-year FY25).
Note: Framework ceilings (\$24.4B) are not funded appropriations.

Known-By Signals

- [] **FTM-43 Live Test:** Will prove if hypersonic defense moves from “simulated” (Rung 3) to “live intercept” (Rung 4).
- [] **Block IB Resolution:** Will indicate if the Navy **restarts** the **paused** kinematic/motor upgrade or relies entirely on software paths.
- [] **FY2027 Appropriations:** Watch for the **funding** of the requested **540 rounds** to gauge true industrial base throughput capacity.